Aaron Emmanuel

Mechatronics | Robotics & Systems Integration

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TECHNICAL EXPERIENCE

Associate Analyst | Rotational Engineering Program *May 2025 - Present*

Kinectrics

- **Inspection and Maintenance Systems (Sep 2026 - Present):** Coordinate component needs with project teams, vendors and machine shops; assemble and functionally test purchased parts before final integration, including work on Calandria Relief System Modifications.
- **Technology and Strategic Ventures (Jan - Sep 2026):** Led proposals and coordinated technical input for radioisotope power and heating applications, contributing to Kinectrics' first Canadian Space Agency partnership.
- **Design and Engineering Services (May 2025 - Jan 2026):** Performed mechanical design analyses and validated electrical/instrumentation-and-control requirements to support qualification of replacement motors, relief valves and actuators at Bruce A and B nuclear stations.

Technical Lead | Systems Design and Integration *Jun 2024 - Present*

Circulr

- Designed, built, programmed and deployed a compact reverse-vending machine for reusable jars and beverage cans as Circulr's sole technical contributor, bringing the system into store testing at Fiesta Farms.
- Integrated mechanical handling, electrical assembly and Raspberry Pi/Python controls with barcode validation, refund values and backend records. Processed approximately one item per second across six jar types and beverage cans, correctly accepting or rejecting every container in the evaluated test set.
- Designed a sensor-controlled storage elevator that moved with fill level to limit glass-container drop height and increase storage capacity.
- Enabled automatic recovery from momentary power loss and Raspberry Pi freezes by implementing boot-time autostart, watchdog monitoring and automatic application restart; the machine remained in service with no reported failures for six months after March 2026.

Co-Founder and Systems Lead | Underwater Robotics *Apr 2026 - Present*

MAKO Technologies

- Co-founded MAKO Technologies to develop a clean-sheet underwater inspection crawler for Clearwater, with scope spanning mechanical, hydraulic, electrical, sensing, controls and software.
- Completed customer requirements and market research; now selecting components and developing a digital model around the competing demands of fitting through a manhole and producing sufficient power and traction to operate against water flow.

Robotic Systems Designer | Research and Development *Jan 2024 - Aug 2025*

Clearwater Structures

- Rebuilt the electronics, control software and communications of an inoperable underwater inspection crawler, restoring full-system operation and demonstrating it underwater at a customer job site.
- Developed custom Python controller software linking onboard and truck-based Raspberry Pis by local IP over Ethernet/fibre, enabling remote operation and automatic stopping on communication loss; integrated a physical emergency stop.
- Integrated pan-tilt-zoom CCTV, LiDAR and an IMU into the rebuilt control system to expand inspection visibility and motion sensing.

SELECTED TECHNICAL PROJECTS

Autonomous Inspection | ROS 2 and Unitree Go2 *Oct 2026*

Independent robotics project | TurtleBot 4 / Gazebo and Go2 / MuJoCo simulation

- Built a ROS 2 Humble inspection application on Ubuntu, progressing from TurtleBot 4 LiDAR SLAM in Gazebo Fortress to AMCL localization, Nav2 path planning and camera-based station checks.
- Connected ROS 2 navigation commands to the official Unitree SDK2 and a custom Sport API simulation server, driving Go2 physics in MuJoCo with a reused walking policy and DDS state feedback.
- Trained a PyTorch CNN to detect cones and fire extinguishers; ONNX evaluation on 600 held-out synthetic images achieved 91.2% precision and 90.2% recall at IoU 0.50.
- Fused RGB detections, aligned metric depth and stamped TF to estimate visible-surface range and maintain six stationary object tracks through occlusion and reacquisition.
- Implemented a C++ Nav2 arrival controller to remove repeated turn reversals; verified inspection completion, newly introduced obstacle detours and blocked-route stopping with logged simulation tests.

Collaborative Autonomous Robot Delivery System *2024 - 2025*

Ontario Tech University | Undergraduate capstone

- Wrote and physically tuned a PID balance controller using IMU feedback and ODrive motor telemetry for two collaborative delivery robots.
- Integrated Jetson, Teensy, ODrive and CAN controls; assembled and tested the robots, including a spring-assisted mechanism that enabled standing.

ASL Recognition Glove | Wearable Sensing *Dec 2025*

University of Toronto | MEng course project

- Delivered live recognition of all 26 ASL letters in two weeks using ESP32/Bluetooth, five flex sensors and MPU6050 acceleration inputs, with webpage and spoken output.
- Collected a two-user dataset, evaluated four classifiers and selected K-nearest neighbours; achieved 99.6% held-out accuracy with an 80/20 train/test split.

TECHNICAL SKILLS

Programming and controls: Python, C++ (ROS 2/SDK integration), embedded programming, PID control.

Robotics and simulation: ROS 2 Humble, Ubuntu/Linux, Unitree SDK2, Nav2, SLAM, AMCL, TF, Gazebo, MuJoCo.

Vision and sensing: OpenCV, PyTorch, ONNX, RGB-D fusion, object detection and stationary-object tracking.

Embedded hardware and interfaces: Raspberry Pi, ESP32, Jetson, Teensy, ODrive, CAN, Ethernet/fibre and Bluetooth.

CAD and electronics design: SolidWorks, Siemens NX, Autodesk Fusion, KiCad; drawings, structural FEA, motion simulation and PCB design.

Fabrication and test: Sensor integration, wiring, soldering, crimping, 3D printing, multimeter/oscilloscope testing and field troubleshooting.

EDUCATION

Master of Engineering | Mechanical and Industrial Engineering *2025 - 2027*

University of Toronto

- Robotics Emphasis; degree in progress.

Bachelor of Engineering | Mechatronics Engineering *2020 - 2025*

Ontario Tech University

- Graduated with Highest Distinction.